A FINITELY PRESENTED MODULE FLAT AT A POINT IS FREE THERE EXACTLY WHEN ITS FIBRE MODULE IS FREE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5939-5957 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5959-5986, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Finitely presented modules flat over a morphism, the local freeness criterion, and etale neighbourhoods; the descent is retained.

Source reference. Original lemma lemma-flat-on-fibres-at-point: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5632–5662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-flat-and-free-at-point-fibre: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5986–5986. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-lemma-describe-etale-local-ring: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5986–5986. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-lemma-stalk-quasi-coherent: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5986–5986. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite presentation. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module. Let $x \in |X|$ with image $y \in |Y|$. If $\mathcal{F}$ is flat at x over Y, then the following are equivalent*

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(1) $(\mathcal{F}_{\overline{y}})_{\overline{x}}$ *is a flat* $\mathcal{O}_{X_{\overline{y}},\overline{x}}$*-module,*
(2) $(\mathcal{F}_{\overline{y}})_{\overline{x}}$ *is a free* $\mathcal{O}_{X_{\overline{y}},\overline{x}}$*-module,*
(3) $\mathcal{F}_{\overline{y}}$ *is finite free in an étale neighbourhood of* $\overline{x}$ *in* $X_{\overline{y}}$*, and*
(4) $\mathcal{F}$ *is finite free in an étale neighbourhood of* x *in* X*.*

Here $\overline{x}$ *is a geometric point of* X *lying over* x *and* $\overline{y} = f \circ \overline{x}$*.*

Proof. Pick a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

where U and V are schemes and the vertical arrows are étale such that there is a point $u \in U$ mapping to x. Let $v \in V$ be the image of u. Applying Lemma lemma-flat-on-fibres-at-point to id : $X \to X$ over Y we see that (1) translates into the condition "$\mathcal{F}|_{U_v}$ is flat over U_v at u". In other words, (1) is equivalent to $(\mathcal{F}|_{U_v})_u$ being a flat $\mathcal{O}_{U_v,u}$-module. By the case of schemes (More on Morphisms, Lemma more-morphisms-lemma-flat-and-free-at-point-fibre), we find that this implies that $\mathcal{F}|_U$ is finite free in an open neighbourhood of u. In this way we see that (1) implies (4). The implications (4) $\Rightarrow$ (3) and (2) $\Rightarrow$ (1) are immediate. For the implication (3) $\Rightarrow$ (2) use the description of local rings and stalks in Properties of Spaces, Lemmas spaces-properties-lemma-describe-etale-local-ring and spaces-properties-lemma-stalk-quasi-coherent. $\square$